Total Environment Research Themes 8 (2023) 100072

Contents lists available at ScienceDirect

Total Environment Research Themes

journal homepage: www.elsevier.com/locate/totert

Soil erosion and sediment yield modeling in a tropical mountain watershed of the southern Western Ghats, India using RUSLE and Geospatial tools

A.L. Achu [a,*], Jobin Thomas [b]

[a] Department of Climate Variability and Aquatic Ecosystems, Kerala University of Fisheries and Ocean Studies (KUFOS), Kochi 682 508, Kerala, India
[b] Department of Geological and Mining Engineering and Sciences, Michigan Technological University, Houghton, MI 49931, USA

ARTICLE INFO

Keywords:
Soil erosion
Sediment yield
RUSLE
SEDD
Karimpuzha
Western Ghats

ABSTRACT

The development of comprehensive soil conservation strategies, based on the soil erosion risk, is one of the critical components of sustainable river basin management, especially in tropical mountainous regions. The present study computes the spatially distributed longtime average annual soil loss (A) and sediment yield (SY) and prioritizes the soil erosion risk in a tropical mountain river basin. The Revised Universal Soil Loss Equation (RUSLE) and the sediment delivery distributed model (SEDD) models were coupled in the GIS environment to predict the intra-basin variability of the A and SY of the Karimpuzha River Basin (KRB) of the southern Western Ghats (in Kerala, India). The results indicate that, on an average basis, 9.3 t of soil has been eroded per hectare per year, whereas the SY is 6.3 t ha^{-1} y^{-1} (i.e., approximately two-thirds of the gross soil erosion). The SY of the majority (74.2%) of the basin area is low (i.e., ≤5.0 t ha^{-1} y^{-1}), whereas the soil erosion and transport processes are a major concern in some areas (ca. 6.3 km^2). The results imply the significance of land use/land cover, rainfall, and topography in controlling the soil erosion risk of the basin. The study demonstrates the advantages of the delineation of the critical erosion areas, based on the hydrologic response units (HRUs) of the basin, to implement appropriate soil erosion management options for conservation of the soil resources.

Introduction

Soil is one of the major natural resources supporting life on the Earth. Soil erosion is a geomorphic process of detaching and transporting soil particles from their origin by the agents of erosion, such as wind and water. Although soil erosion is a natural process, its intensity is exacerbated by various anthropogenic activities (Fayas et al., 2019; Othman et al., 2023). Steep and rugged topography, unscientific farming practices, and overgrazing are some of the critical factors influencing the severity of soil erosion (Singh & Kansal, 2023). Accelerated erosion of fertile topsoil along with available nutrients is integrally linked to land degradation (Li et al., 2009; Porto et al., 2022). Recent studies estimated that soil erosion facilitated by water is about 20–30 Gt per year, while tillage erosion accounts for about 5 Gt annually (FAO & ITPS, 2015). Considering the gravity of the extent and severity of land degradation globally, the Sustainable Development Goal (SDG) 15 seeks to protect, restore, and promote the conservation and sustainable use of terrestrial, inland water, and mountain ecosystems. Besides, SDG 15 calls for implementing measures to combat desertification and restore degraded land and soil to achieve a land degradation-neutral world by 2030 (https://sdgs.un.org).

The healthy river basins offer social, environmental, and economic security by providing various goods and services, including nutrient cycling, carbon storage, and erosion/sedimentation control. However, the accelerated soil erosion, guiding to improper functioning of the ecosystem services, needs to be quantified at the watershed scale to design appropriate soil erosion management strategies. Modeling of soil erosion provides quantitative estimates of soil loss across a region under varying environmental conditions. A wide range of soil erosion and sediment yield models have been developed for stimulating the soil erosion rates and sediment yield at different spatial and temporal scales. The common approaches in soil erosion modeling involve the use of either empirical models, such as the Universal Soil Loss Equation (USLE) (Wischmeier and Smith, 1978), MUSLE (Williams, 1975), RUSLE (Renard et al., 1997), and USPED (Mitas and Mitasova, 1998), or physically-based models, such as the ANSWERS (Beasley et al., 1980), WEPP (Nearing et al., 1989), and EUROSEM (Morgan et al., 1998). Although the empirical models are characterized by simplified process representations, which is a major limiting factor influencing the model accuracy, the Universal Soil Loss Equation (USLE) and its revised form,

* Corresponding author.
E-mail address: achu.geomatics@gmail.com (A.L. Achu).

https://doi.org/10.1016/j.totert.2023.100072
Received 27 January 2023; Received in revised form 9 July 2023; Accepted 5 August 2023
Available online 11 August 2023
2772-8099/© 2023 The Author(s). Published by Elsevier B.V. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

the Revised Universal Soil Loss Equation (RUSLE) have been the most widely used empirical models for long-term, quantitative soil loss estimation because of its minimal data requirements and compatibility with geospatial platforms (Millward and Mersey, 1999; Aiello et al., 2015; Masroor et al., 2022; Saha et al., 2022; Avand et al., 2023). Since the RULSE does not account for sediment transport and redistribution, overland flow connectivity, and deposition of the eroded sediments, the sediment delivery distributed model (SEDD) is effectively used in conjunction with the RUSLE to predict the net soil erosion rates and to compute the sediment yield at the catchment scale (e.g., Tanyaş et al., 2015; Thomas et al., 2018a; Mirakhorlo and Rahimzadegan, 2020; Batista et al., 2021; Kanito et al., 2023).

The Western Ghats is an important geomorphic feature bordering the western edge of the Indian Peninsula and facilitates unique orographically-driven elevation and bio-climatic gradients and thresholds to yield distinctive climates between the windward and leeward slopes. The Western Ghats is also a major hotspot of soil erosion in India due to the unique combination of topographic, climatic, and socioeconomic factors. Although Milliman & Syvitski (1992) emphasized the significance of small mountain rivers in global sediment influx into the oceans, catchment erosion and sediment generation in the mountain rivers of the Western Ghats are less explored. A few researchers (e.g., Prasannakumar et al., 2012; Thomas et al., 2018a, b, c; Sujatha and Sridhar, 2019; Chinnasamy and Honap, 2023) demonstrated the significance of the unique combination of rainfall patterns mediated by the topography and land use/ land cover on soil erosion rates in a few watersheds across the region. However, most of the studies focused on the severity of soil erosion in the watersheds of the Western Ghats south of the Palghat Gap (particularly in the Anaimalai-Cardamom Hills). Despite the comparable physiographic and climatic conditions of the Anaimalai-Cardamom Hills, soil erosion in the watersheds draining the Nigiri Massif is poorly addressed. In this study, we estimate the long-term soil erosion rates and sediment yield in the Karimpuzha River Basin (KRB) draining the Nilgiris of the Western Ghats (Kerala, India) using the RUSLE and SEDD models. Elsewhere across the Western Ghats, soil erosion is a major concern in the KRB. The KRB has been subjected to remarkable land use/ land cover changes during the colonial period and the region witnessed a transition from natural vegetation to commercial plantations. The Nilambur teak plantations, extending across the KRB, are one of the oldest Teak plantations worldwide. In addition, the extreme storm events occurring during the Indian summer monsoon season aggravate the problems associated with soil erosion.

Materials and methods

2.1. Study area

The Karimpuzha River Basin (KRB, area = 313 km^2) is a major sub-basin of the Chaliyar River (Fig. 1). The KRB drains the western slopes of the Western Ghats between the latitudes of 11° 14′ 56″ N and 11° 23′ 55″ N and longitudes of 76° 15′ 37″ E to 76° 34′ 38″ E. The Karimpuzha River originates from the Porthimund peak at Nilgiri hills (2577 m above the mean sea level) and joins with other major tributaries of the Chaliyar River at the Nilambur Valley, a major geomorphic feature of the region. The KRB experiences a tropical monsoon climate which is characterized by two distinct rainfall seasons, viz., the southwest monsoon or Indian summer monsoon (June to September) and the northeast monsoon (October to December), which together contribute about 90% of the

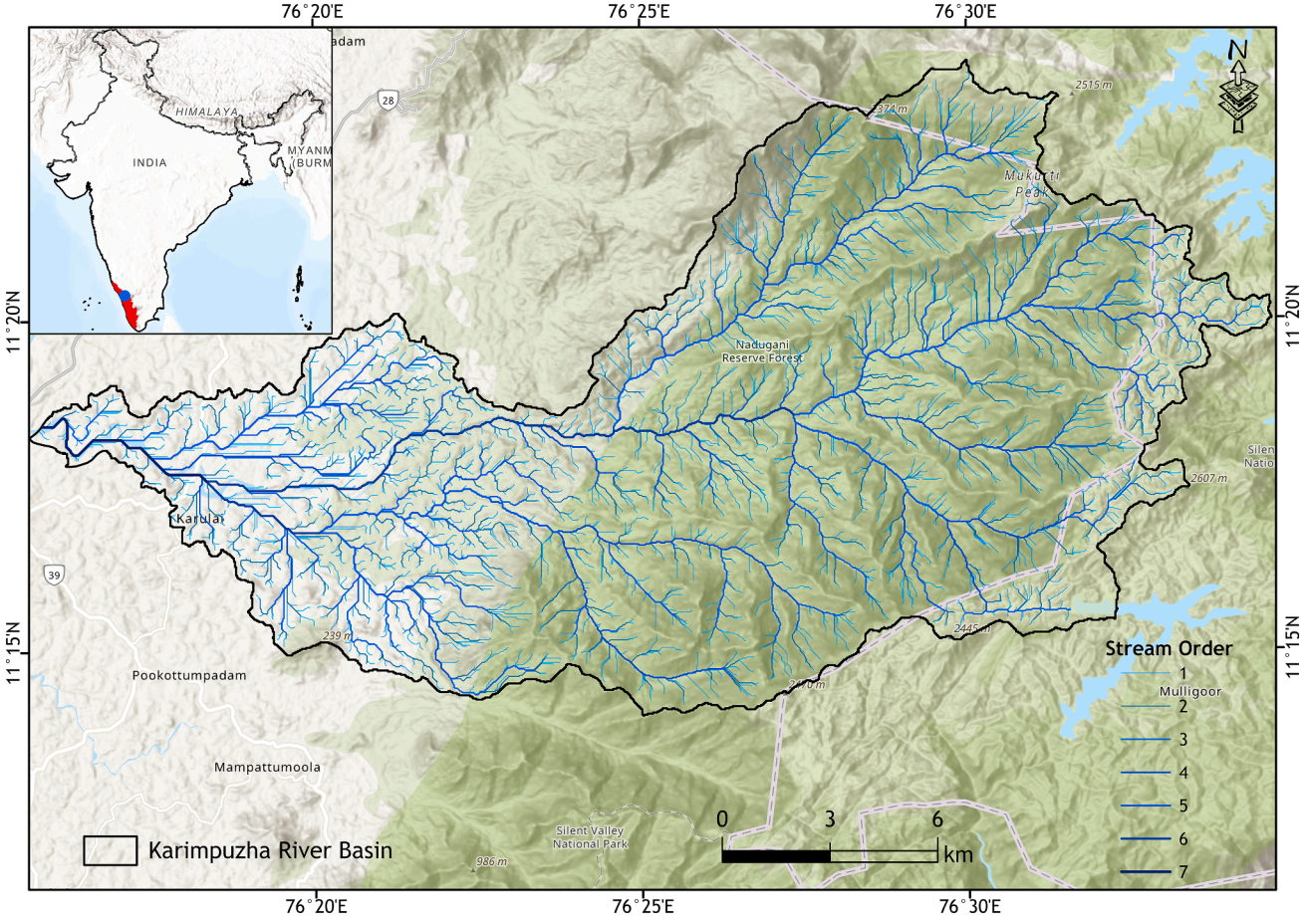


Fig. 1. Location map of the Karimpuzha River Basin with stream network.

annual rainfall budget. The mean annual rainfall of the basin ranges from 1598 to 2445 mm, as recorded by different meteorological stations of the region with a mean annual temperature of about 30℃ (Source: India Meteorological Department, 2006–2020). The study area forms part of the Precambrian metamorphic shield with rocks of the Wayanad Group comprising of peninsular gneissic complex, charnockite group of rocks, high-grade meta-sedimentary rocks, and intrusive migmatite complex (Soman, 2002; Ambili and Narayana, 2014).

2.2. RUSLE model

The RUSLE model (Renard et al., 1991) has been a widely used empirical soil erosion model to predict the average annual soil loss in different environmental settings including agricultural and forested landscapes. In the present study, we considered only the overland areas and excluded the stream channel and waterbodies from the analysis. The stream channels were discriminated from the overland areas using an appropriate channel initiation threshold (Wang and Yin, 1998; Thomas and Prasannakumar, 2015). In the RUSLE, the soil loss due to water erosion per unit area per year is computed as Eq. (1) (Renard et al. 1997).

$$A = R \times K \times L \times S \times C \times P \tag{1}$$

where A is the computed spatial average of soil loss over a period (in t/h y^{-1}), R is the rainfall-runoff erosivity factor (MJ mm ha^{-1} h^{-1} y^{-1}), K is the soil erodibility factor (t ha h/ha MJ^{-1} mm^{-1}), L is the slope length factor (dimensionless), S is the slope steepness factor (dimensionless), C is the cover and management factor (dimensionless) and P is the conservation support-practices factor (dimensionless). The spatially distributed input factors of the RUSLE model (Eq. (1)) were generated in ArcGIS to predict the A of the basin. Details of the estimation of the RUSLE input variables are given in the following sections.

2.2.1. Rainfall erosivity factor (R)

Rainfall and corresponding runoff are the significant drivers of sheet and rill erosion processes, and hence, the R factor explains the effect of raindrop impact as well as the amount and rate of runoff associated with the rain. The R factor is considered the most correlated variable to soil loss (Hlaing et al., 2008; Ouadja et al., 2022). In this study, we used the modified Fournier index (F) (Eq. (2)) developed by Arnoldus (1977, 1980) for the estimation of the R factor (MJ mm ha^{-1} h^{-1} y^{-1}) of different meteorological stations in and around KRB.

$$F = \frac{\sum_{i=1}^{12} p_i^2}{P} \tag{2}$$

where p_i is the monthly rainfall (mm) and P is the annual rainfall (mm). The monthly and annual rainfall data of four meteorological stations were collected from India Meteorological Department for the period from 2006 to 2020, used to derive R. The spatially distributed R factor of the KRB was generated using the probabilistic inverse distance weighted (IDW) interpolation method.

2.2.2. Soil erodibility factor (K)

The soil erodibility is a function of soil texture, structure, organic matter content, and permeability (Wischmeier et al., 1971). The K factor represents the average long-term response of soil and soil profile to a number of erosion and hydrological processes, such as soil detachment, transportation due to the impact of the raindrop and overland flow shear forces, localized deposition due to topography and tillage-induced roughness, and rainwater infiltration into the soil profile. Hence, the K factor is marginally interdependent on the R, LS, and C factors of the RUSLE (Borah et al., 2008). Being a forest-dominated landscape, detailed soil characteristics for the calculation of the K factor are not available. Therefore, the available soil textural data of the region were collected from the Soil Survey Organization, Government of Kerala (SSO, 2007), and computed the K factor. The dominant soil texture in the KRB is clay (area = 53.2%), followed by gravelly clay (21.9%), silty clay (13.8%), and silty loam (11.1%) (Table 1). The K values of each soil type were identified from the soil erodibility nomograph (USDA 1978) by considering the particle size, organic matter content, and permeability class.

2.2.3. Slope-length and steepness factor (LS)

The LS factor is one of the critical factors in the RUSLE model that influences soil erosion, transportation, and sediment yield (Wischmeier and Smith, 1978). The effect of topography on soil erosion is represented by two variables, viz., slope length (L) and slope steepness (S). We used Eq. (3), based on unit stream power theory, to derive the combined LS factor (Moore and Burch 1986a,b; Moore and Wilson 1992).

$$LS = \left[\frac{A_{si}}{22.13}\right]^n \left[\frac{sin\alpha_i}{0.0896}\right]^m \tag{3}$$

where A_{si} is the specific catchment area of cell i, defined as the upslope contributing area per unit width normal to the flow direction and α_i is the slope gradient of cell i in degrees. We used the optimal parameter values of m and n in Moore and Wilson (1992) for natural landscapes, i.e., 0.6 and 1.3, respectively. The SRTM DEM (1 arc second) was used to generate the LS factor of the KRB.

2.2.4. Cover and management factor (C)

The vegetation cover is, after topography, the second most influential factor controlling soil erosion (van der Knijff et al., 2000). The C factor is defined as the ratio of soil loss from the land cropped under specific conditions to the corresponding loss from the clean-tilled, continuous fallow (Wischmeier and Smith, 1978). In this study, we estimated the C factor from the satellite-derived normalized difference vegetation index (NDVI), which is an indicator of vegetation vigor, using Eq. (4).

$$C = \exp\left[-\alpha \times \frac{NDVI}{\beta - NDVI}\right] \tag{4}$$

where α and β are the unitless parameters defining the shape of the NDVI and C-factor curve. Van der Knijff et al. (2000) noted reasonable accuracy for the C factor while using the α as 2 and β value of 1 and hence, we also used the parameter values invariably. The NDVI was derived from the Landsat 8 OLI image.

2.2.5. Conservation practice factor (P)

The P factor is an expression of the overall effects of various soil conservation measures (e.g., contouring, strip-cropping, terracing, and subsurface drainage) on soil erosion, as the support practices affect the soil erosion by water through modification of the flow pattern, grade, or direction of surface runoff, as well as through reduction of the rate and volume of runoff (Wischmeier and Smith, 1978; Renard and Foster, 1983). In this study, the P factor of the basin was estimated by assessing the land use/land cover types that were generated using the Landsat 8 OLI image. We used random field visits to identify the support practices adopted in each land use/land cover type. We assumed that natural vegetation types do not have support practices (and hence a P factor of 1). Since the land use/ land cover types, such as barren rock, rocky cliff/

Table 1
Soil properties of the KRB.

Texture	Local soil series	Physiography[#]	K factor	Area (km^2)
Clay	Uppanmala	Highland	0.2	166.6
Gravelly clay	Vazhikadavu	Midland	0.1	68.5
Silty clay	Vaniampuzha	Highland	0.3	43.3
Silty loam	Anakkayam	Midland	0.4	34.6

[#] Physiographic classification provided in SSO (2007).

escarpment, and sheetrock do not have any soil cover to be removed, we considered the soil loss in such land use/ land cover types as zero. All the RUSLE input factors were generated at a spatial resolution of 30 m using UTM Zone 43 N projection.

2.2.6. Sediment delivery distributed (SEDD) model and sediment yield (SY)

The entire volume of the eroded soil does not reach the stream channel cells as a large portion of the eroded soil will be redistributed and deposited within the watershed. The ratio of SY to gross soil erosion (i.e., A) in the basin is defined as the sediment delivery ratio (SDR). The spatial variability of the SDR in a catchment is controlled by numerous catchment characteristics including physiography, sediment sources, sediment connectivity, and vegetation cover (Richard, 1993). Ferro and Minacapilli (1995) and Ferro et al. (1998) represented the SDR as a function of travel time of the overland flow (Eq. (5)), which is influenced by the topography and vegetation cover.

$$SDR_i = \exp(-\gamma t_i) \tag{5}$$

where t_i is the travel time (h) for cell i to the nearest channel cell and γ is a watershed-specific parameter, implying the basin morphology. Fernandez et al. (2003), Jain and Kothyari (2000) and Fu et al. (2006) assessed the sensitivity of γ on sediment yield and reported that sediment yield is not very sensitive to γ (0.1–2.0). In this study, we used the value of 1 for γ as we lack observed data for calibration.

The travel time from a cell to the nearest channel cell is a function of the flow distance and velocity of the flow path. If the flow path from cell i to the nearest channel traverses through N cells, then the travel time is calculated as the sum of the travel time for each of the N cells along the flow path (Eq. (6)).

$$t_i = \sum_{i=1}^{N} \frac{li}{vi} \tag{6}$$

where l_i is the length of segment i in the flow path (m) and v_i is the flow velocity in the cell (m/s). We used the method proposed by US Soil Conservation Service to compute the overland flow velocity as a function of both surface slope and land cover characteristics (SCS, 1975) (Eq. (7)).

$$v_i = c_i S_i^m \tag{7}$$

where m is equal to 0.5 (SCS, 1975; Ferro and Minacapilli, 1995), S_i is the slope of the cell i (m/m) and c_i is a coefficient related to land cover type (after Haan et al., 1994). In addition, a lower limit of velocity is generally considered by setting the minimum cell slope to a small value

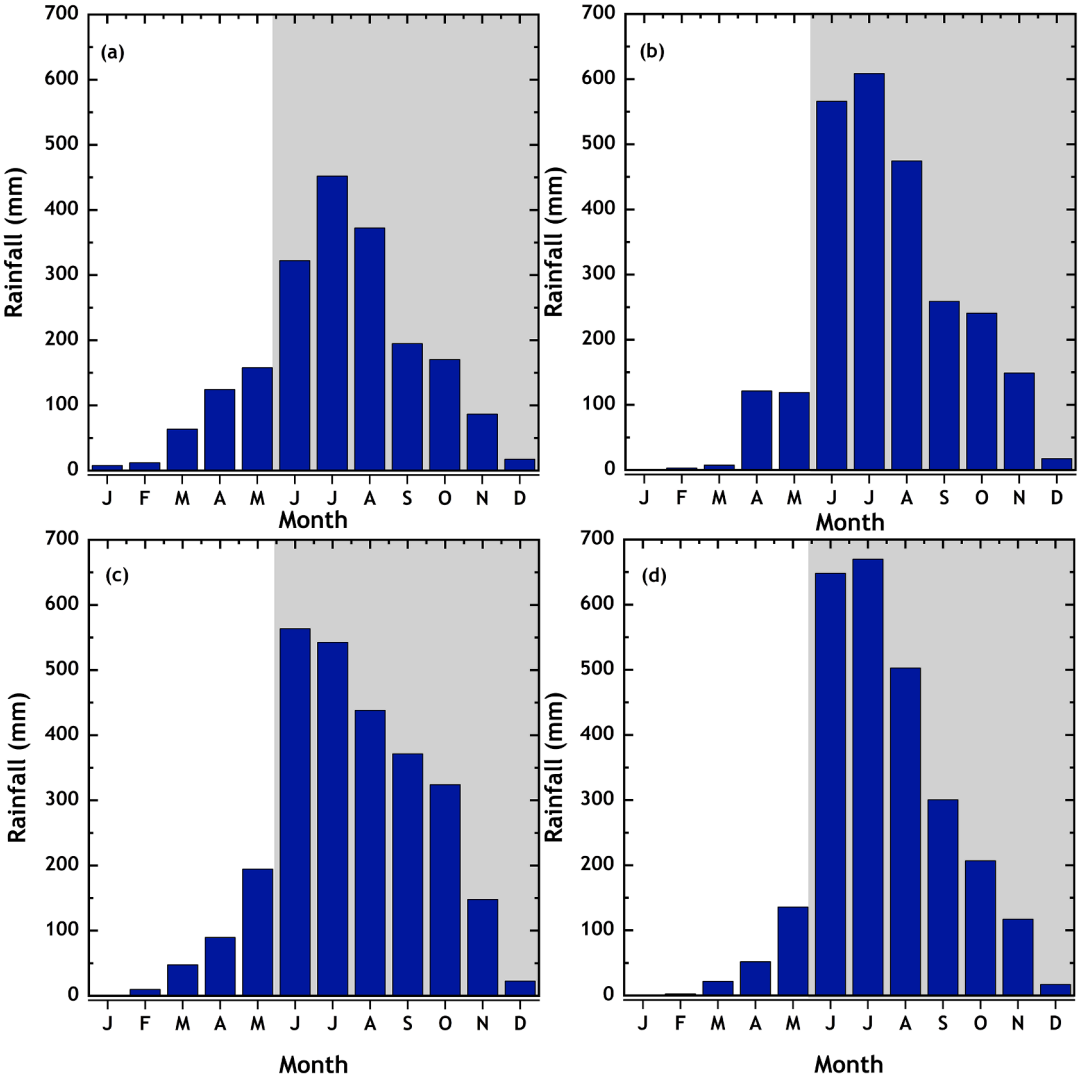


Fig 2. Mean Monthly rainfall of different meteorological stations in KRB (a) Ambalavayal, (b) Manjeri, (c) Mannarkkad, and (d) Nilambur for the period 2006–2020. Grey shaded portion indicates monsoon season (Indian summer monsoon from June to September and North-East monsoon from October to December).

(e.g., 0.3% in this study) (Ferro and Minacapilli, 1995; Fernandez et al., 2003., Fu et al., 2006). If A_i is the amount of long-term annual soil loss in a given cell (derived using Eq. (1)), then the SY is estimated as Eq. (8).

$$SY = SDR_i \times A_i \quad (8)$$

3. Results and discussion

3.1. RUSLE factors

The monthly rainfall recorded in all the stations in and around the KRB shows significant variations and the Indian summer monsoon season spanning from June to September contributes ~90% of the annual rainfall followed by the northeast monsoon season (Fig. 2). Among the four rain gauge stations used in this study, Nilambur and Mannarkad receive the maximum mean annual rainfall (i.e., 2445 and 2387 mm respectively), followed by Manjeri (2298 mm) and Ambalavayal (1598 mm). In the KRB, the R factor ranges between 470.9 and 593.8 MJ mm ha^{-1} h^{-1} y^{-1} (Fig. 3a) and the western part of the KRB has relatively higher R values than the rest of the catchment area. The spatial variability of the R factor is largely contributed by the spatial differences in the rainfall pattern and annual rainfall total. The mean R factor of the basin is 510 MJ mm $ha^{-1}h^{-1}$ y^{-1}, which is comparable with many previous studies in the Western Ghats (e.g., Pradeep et al., 2015; Thomas et al., 2018b,c). The soil texture of the KRB varies from clay to silty loam and the K values of the different soil types range from 0.1 to 0.4 t ha h ha^{-1} MJ^{-1} mm^{-1} with a mean K factor of 0.2 t ha h ha^{-1} MJ^{-1} mm^{-1} (Fig. 3b). The LS factor ranges from 0 to 23.2 with a mean value of 0.7 (Fig. 3c), where high LS values are distributed across the areas with steep slopes. The majority (98%) of the catchment area has LS values less than 5. The LS factor is referenced to unity at 22.1 m slope length and 9% steepness (Wischmeier and Smith, 1978), and hence, the relatively larger LS factor values in the KRB suggest the higher potential of the terrain for soil erosion.

The C factor of the KRB varies from 0.07 to 1.1 with a mean value of 0.6 (Fig. 3d). High C values are characterized in either open scrub areas or rocky exposure. Since NDVI is inversely related to the C factor, low C values are the manifestations of high NDVI values, implying healthy vegetation. The variability of the C factor in the KRB is similar to the range of C factors of different watersheds in the region (Prasannakumar et al. 2011a; Thomas et al., 2018b, c) and Prasannakumar et al. (2011a) observed that the C factor varies between nearly zero (i.e., for well-

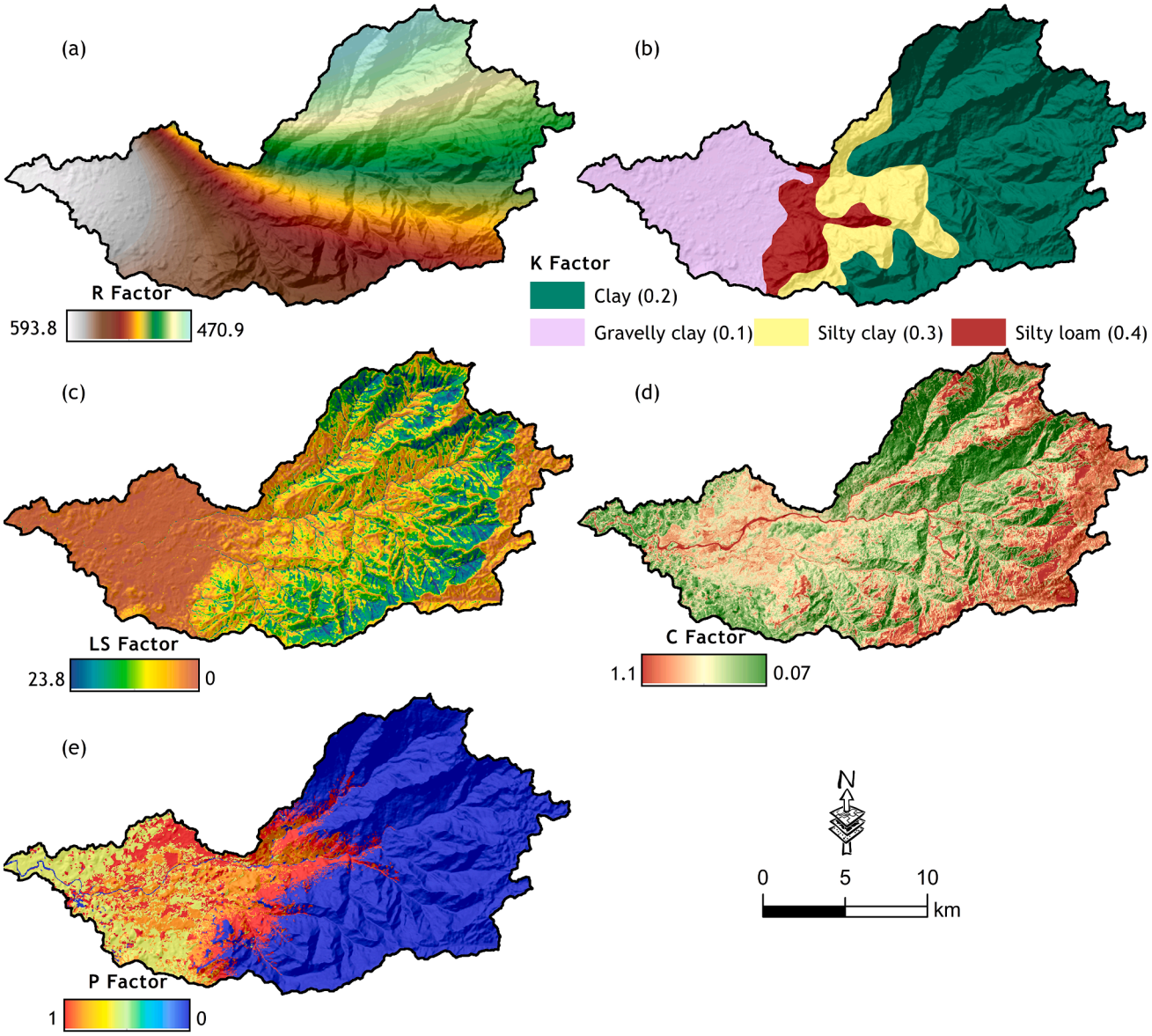


Fig. 3. Spatially distributed RUSLE model inputs of KRB (a) R factor, (b) K factor, (c) LS factor (d) C factor, and (e) P factor.

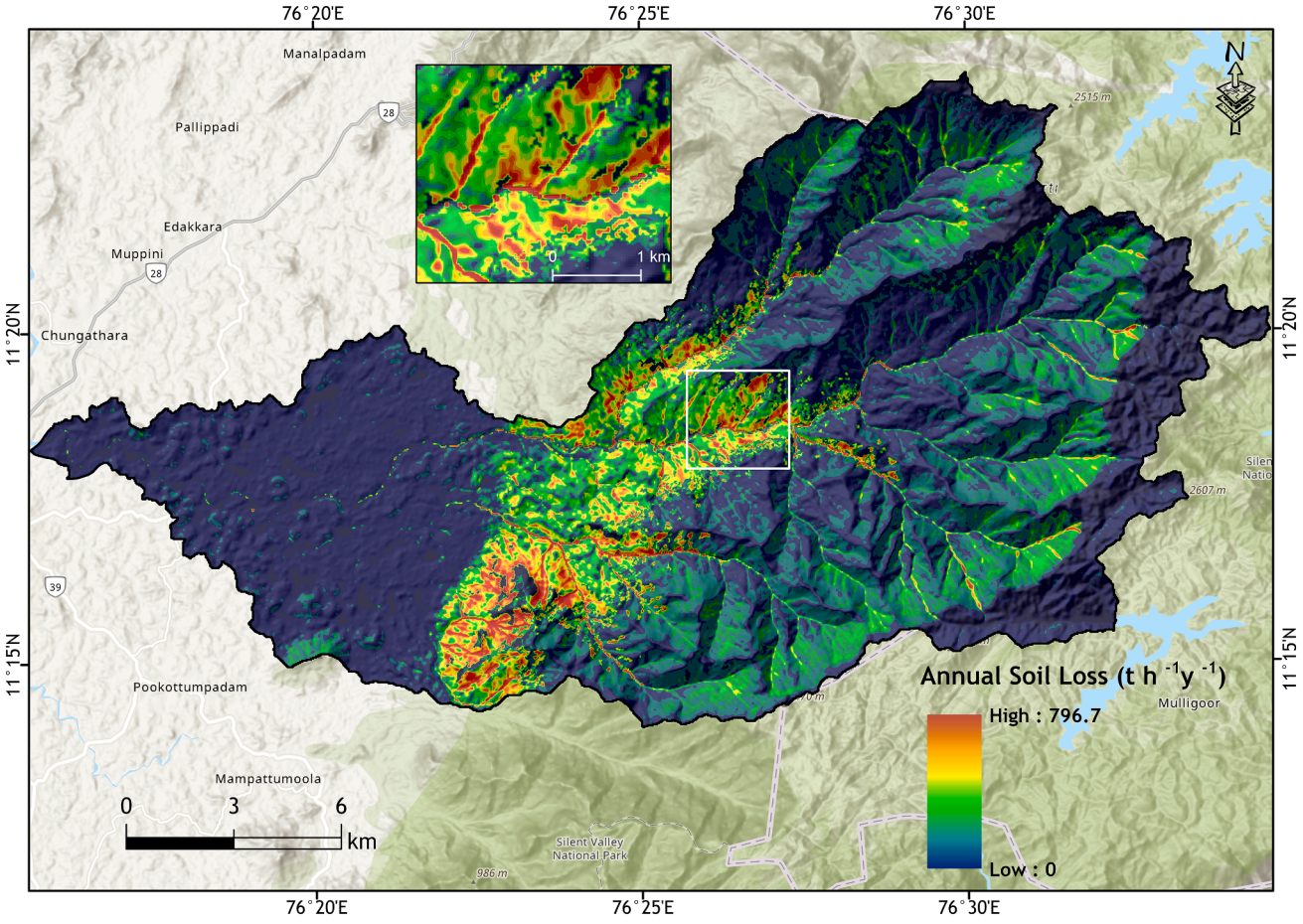


Fig. 4. Long-term annual average soil erosion (A) in KRB predicted by RUSLE model.

protected soils) and 1.5 (i.e., for finely tilled, ridged surfaces). The P factor values of the study area depend on the type of conservation measures adopted in the field to modify the overland flow velocity and pattern. Diverse land use/ land cover types exist in the study area and tropical evergreen forest dominates the basin (~53.6% area) followed by forest plantations (9.5%). Other land use/ land cover types include sheet rocks (8.6%), deciduous forest (7.6%), barren land (7.1%), coconut (6.4%), teak (3.9%), and rubber plantations (2.4%), built-up area (0.3%) and waterbody (0.6%). The natural vegetation types (with no support practices) are characterized by the P factor value of 1 (Fig. 3e).

3.2. Estimation of mean annual gross soil loss

The RUSLE model inputs (i.e., R, K, P, LS, and C) were integrated with ArcGIS to predict the long-term soil loss (A) on a pixel-to-pixel basis (30 m resolution). The predicted A of the KRB ranges from 0 to 796 t h^{-1} y^{-1} with a mean gross soil loss of 9.3 t h^{-1} y^{-1} (Fig. 4). In order to understand the variation of the severity of soil loss, the study area is classified into areas of different soil erosion intensities after the thresholds suggested by Singh et al. (1992) for Indian conditions. The soil erosion classes of the KRB suggest that the majority area (51.4%) of the basin experiences slight erosion (i.e., <5 t h^{-1} y^{-1}), whereas only 0.9% of the area is under very severe soil erosion intensity (Table 2). The spatial pattern of predicted A shows that the severe and very severe soil erosion classes are largely distributed across the middle portion of the basin. These regions are characterized by a higher R factor of 552 MJ mm ha^{-1} h^{-1} y^{-1}, K value of 0.3 t ha h ha^{-1} MJ^{-1} mm^{-1}, LS value of 20.1 C factor of 1, and P value of 0.9, shows an increasing trend with soil erosion.

The predicted soil loss of KRB is comparable with various soil erosion studies conducted across different watersheds in the southern Western Ghats region (Table 3). In comparison with other studies from the Western Ghats (e.g., Prasannakumar et al., 2011a, b, 2012; Pradeep et al., 2015; Thomas et al., 2018b, c), the KRB has a relatively lower soil erosion rate. Prasannakumar et al. (2011a) reported a relatively higher rate of soil erosion in the Munnar forest division (24.76 t h^{-1} y^{-1}), primarily due to higher rainfall, rugged topography as well as lack of conservation practices. Contrastingly, the predicted lower erosion rate in a sub-basin of the Pamba River (maximum erosion rate up to 17.73 t h^{-1} y^{-1}) is significantly lower than the KRB (Prasannakumar et al., 2012). Thomas et al. (2018b) reported a mean A of 14.36 (t/h y^{-1}) for the Muthirapuzha River Basin, where the mean annual rainfall is 3700 mm. However, Pradeep et al. (2015) reported a higher rate of soil loss (200.15 t h^{-1} y^{-1}) for a sub-basin of Meenachil River, where mean annual rainfall ranges from 3000 to 3200 mm with rubber plantation as the dominant land use/land cover type.

The soil erosion rate predicted by the RUSLE model in the KRB is also

Table 2
Severity of soil erosion in the KRB.

Severity	Soil erosion (t/h y^{-1})	Area	
		km^2	%
Slight	≤5.0	147.1	51.4
Moderate	5.1–10.0	70.6	24.7
High	10.1–20.0	36.4	12.7
Very high	20.1–40.0	18.6	6.5
Severe	40.1–60.0	10.6	3.7
Very severe	≥60.1	2.6	0.9

Table 3
Comparison of the soil erosion rate of KRB with other studies from the southern Western Ghats.

Study area	Rainfall (mm)	Soil loss (t/h y^{-1})	Reference
Karimpuzha (Chaliyar)	2440	9.3	This Study
Muthirapuzha (Periyar)	3700	14.36	Thomas et al., (2018b)
Pambar	1533	11.70	Thomas et al., (2018c)
Kalathukadavu and Poonjar (Meenachil)	3000–3200	200.15	Pradeep et al., (2015)
Pamba Ar (Pamba)	3046	17.73	Prasannakumar et al., (2012)
Munnar Forest Division	2790	24.76	Prasannakumar et al., (2011a)
Siruvani (Bhavani)	1061	0–14.92	Prasannakumar et al., (2011b)

compared with different studies from other parts of India and across the globe. Masroor et al. (2022) modeled long-term annual soil loss of a subbasin of the Godavari River, where the soil erosion rate is 9.88 t h^{-1} y^{-1}. Serbaji et al. (2023) modeled the long-term annual soil loss of Tunisia, where the annual soil loss shows significant spatial variability ranging from 0 to 619.44 t h^{-1} y^{-1} and observed that LS and K factors strongly control the soil erosion process. Sourn et al. (2022) studied the temporal pattern of annual soil loss in Battambang province in Cambodia and noted that the mean annual soil loss was 2.92 t h^{-1} y^{-1}in 1998, 4.20 t h^{-1} y^{-1} in 2008, and 4.98 t h^{-1} y^{-1} in 2018.

3.3. Sediment delivery ratio (SEDD-SDR) and sediment yield

The SDR values of the KRB range from 0.03 to 1.0 with a mean of 0.24. Fig. 5a shows the spatially distributed SEDD-SDR, where the higher SDR values are associated with steep headwater areas. The computed sediment yield of the KRB ranges between 0 and 690 t h^{-1} y^{-1} with a mean yield of 6.3 t h^{-1} y^{-1}. The spatial variation of SEDD-SY in the KRB shows the major sediment source areas (Fig. 5b). The spatially distributed SY is classified based on severity (Table 4), where the majority of the catchment area (~75%) has an SY of 5 t ha^{-1} y^{-1} or less. Although the soil erosion rate of the KRB is comparable with various watersheds across the southern Western Ghats, the SY is comparatively higher compared to the available SY estimates from the region. For instance, Thomas et al. (2018a) reported that the sediment transport

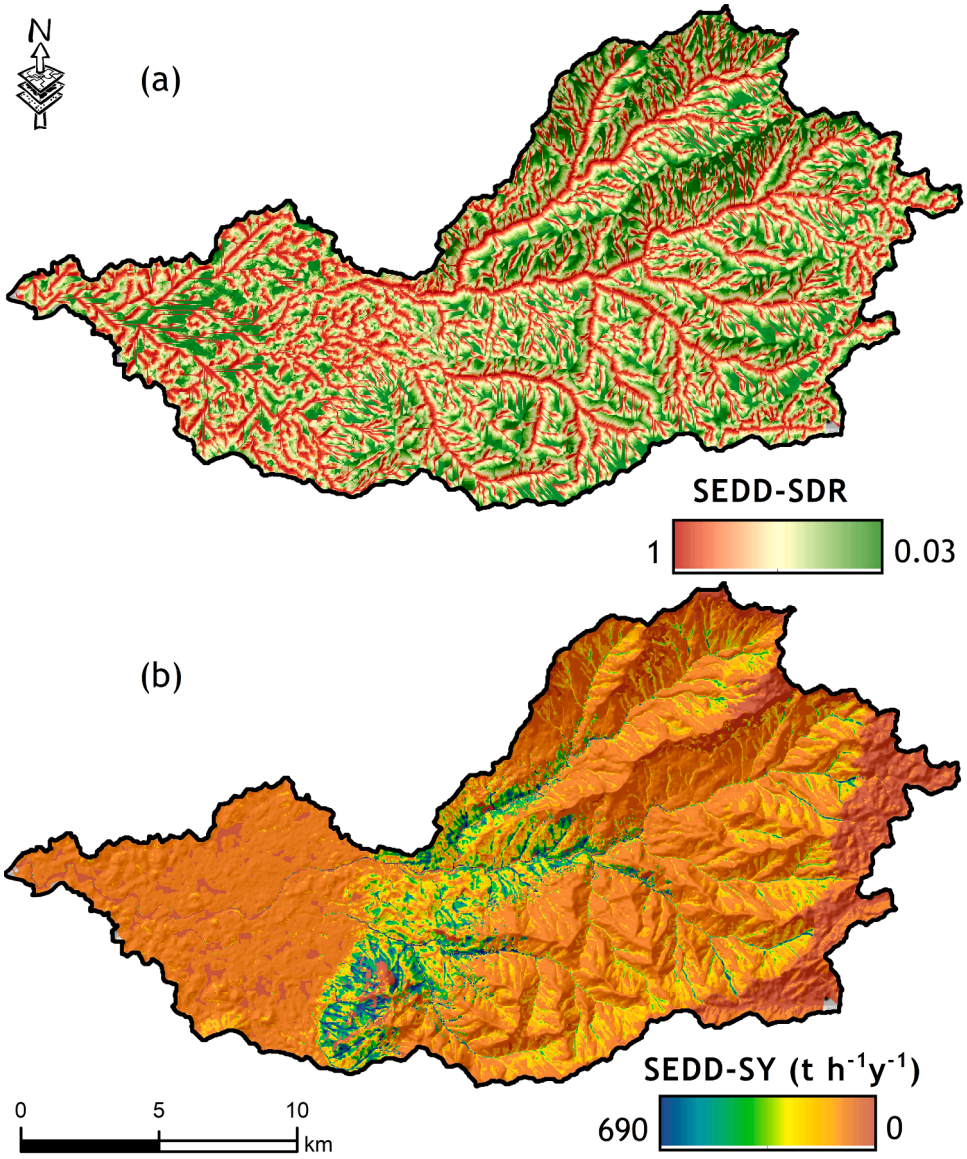


Fig. 5. (a) SEDD-SDR and (b) SEDD-sediment yield variation in KRB.

Table 4
Areal extent of long-term average SY severity classes in KRB.

Severity	SY (t/ha y^{-1})	Area	
		km^2	%
Slight	≤5.0	211.8	74.2
Moderate	5.1–10.0	35.4	12.4
High	10.1–15.0	12.4	4.3
Very high	15.1–20.0	7.0	2.5
Severe	20.1–40.0	12.6	4.4
Very severe	≥40.0	6.3	2.2

efficiency of the Muthirapuzha River Basin is about 25%, whereas, in the KRB, roughly two-thirds of the eroded soil is being transported from the watershed. The higher sediment transport efficiency of the KRB has critical implications for the crop productivity of the region necessitating for framing better watershed management strategies for mitigating soil erosion risk.

3.4. Prioritization of soil erosion risk zones

Although the RUSLE-SEDD approach provides the spatially distributed A and SY, the policymakers and land managers are most interested in the identification and prioritization of critical erosion areas within the basin for implementing soil erosion management strategies. One of the general approaches for prioritization of the critical erosion areas is based on the computed rate of the A or SY, where the mean value of the A or SY of the basin/sub-basin is considered (e.g., Tripathi et al., 2003; Hlaing et al., 2008; Bewket and Teferi, 2009). In such cases, the spatial variability of the A or SY within the basin/sub-basin is ignored. Another common approach is to classify the spatially distributed values of the A or SY into different classes based on user-defined threshold values (e.g., Pandey et al., 2007; Chatterjee et al., 2014; Pradeep et al., 2015), where a range of values is grouped into a single priority class.

The demarcation of the critical areas vulnerable to soil erosion helps determine the effective management options for controlling soil erosion, governed by different natural and anthropogenic factors. Hence, a systematic understanding of the variability of the slope, land use/ land cover, and soil characteristics is a pre-requisite for the implementation of soil erosion management strategies, which is predominantly due to the significance of topography, vegetation, and soil characteristics on the rate of soil erosion of the basin. Hence, the delineation of critical erosion areas based on the hydrologic response units (HRUs) is a better option, where identical soil erosion response areas can be delineated through the grouping of identical hydrologic response areas (Kumar and Mishra, 2015; Haregeweyn et al., 2017).

In this study, the critical erosion areas of the basin were identified based on the HRUs of the basin. The KRB was spatially discretized into 355 HRUs based on the spatial variability of the slope, land use/ land cover, and soil types, and the SY was estimated at the HRU level. Further, the HRUs were categorized into five classes (i.e., very low, low, moderate, high, and very high) based on the SY values (Fig. 6). Among the different classes, both the very low and low priority HRUs together account for about 72% (i.e., 213 HRUs) of the total area of the basin. Other classes, such as moderate, high, and very high-priority HRUs comprise 19.5%, 5.4%, and 2.7 % of the catchment area, respectively (Table 5). The mean A of the high and very high priority zones ranges from 20 to 40 t h^{-1} y^{-1} and 40 to 128.6 t h^{-1} y^{-1}. A total of 71 HRUs are found to have higher vulnerability to soil erosion.

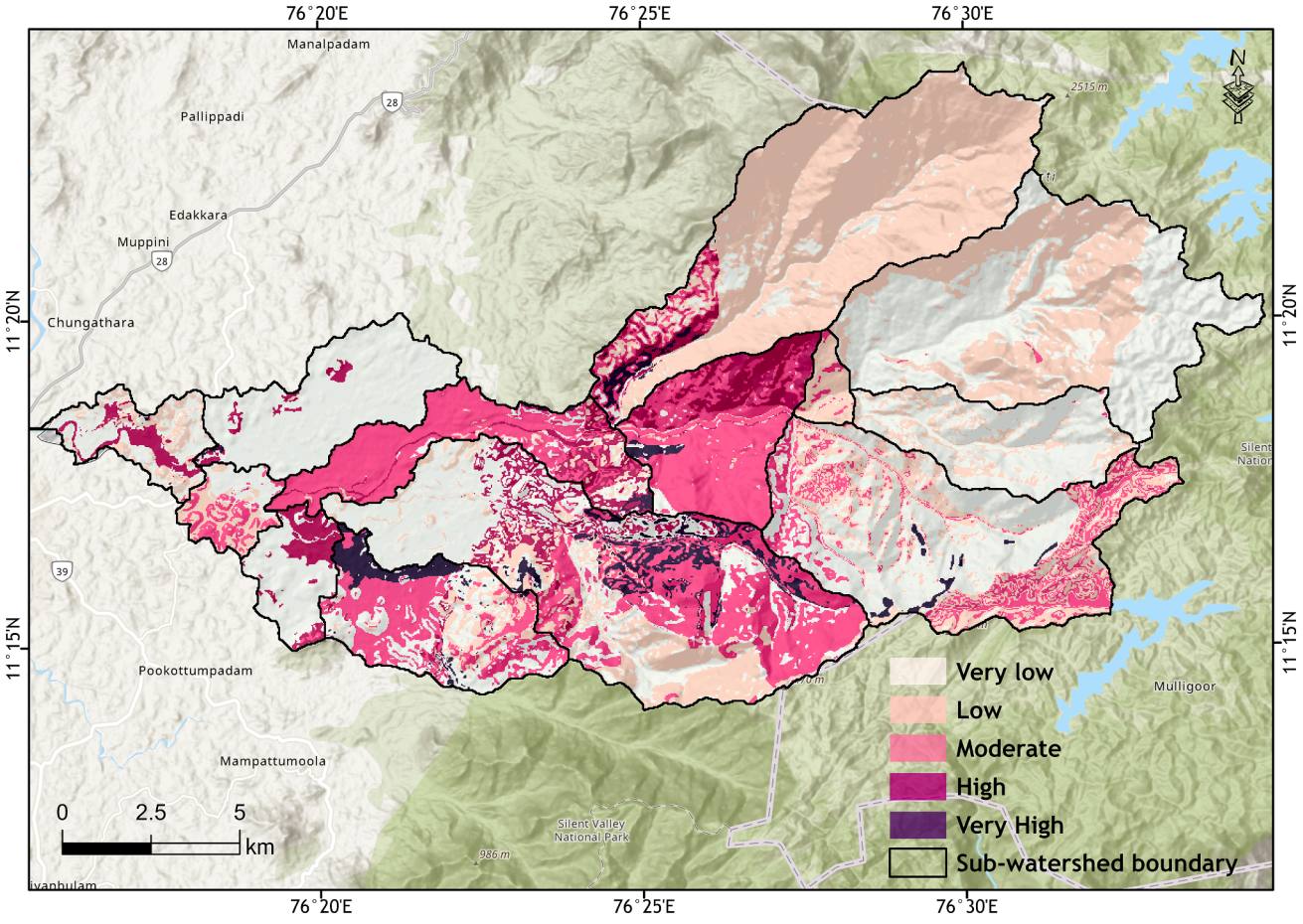


Fig. 6. Spatial distribution of prioritized HRUs across KRB.

Table 5
Prioritization of soil erosion management areas using HRUs.

Priority	Mean Soil loss (t/h y^{-1})	Number of HRUs	Mean slope	Area (km^2)	Area (%)
Very low	0–5	143	22.6	124.9	39.9
Low	5–10	70	30.8	101.5	32.5
Moderate	10–20	71	28.7	60.9	19.5
High	20–40	50	21.6	16.9	5.4
Very high	40–128.6	21	37.3	8.3	2.7

Since HRU is a unique combination of slope, land use/ land cover, and soil type, the spatial variability of the SY with respect to different slope classes, soil types, and land use/ land cover types can be assessed effortlessly. For instance, the very high-priority HRUs are characterized by three major factors: steep slopes (mean slope = 37°), land use/ land cover classes, such as forest plantation and deciduous forest, and areas with higher rainfall. The results suggest that the land use/ land cover, rainfall, and topography play vital roles in controlling the soil erosion risk in the KRB. Hence, it is recommended that the delineation and prioritization of the critical erosion areas based on the HRUs are beneficial for designing cost-effective strategies for soil erosion management as well as for the identification of the dominant factors responsible for soil erosion.

4. Conclusion

The extent and severity of soil loss due to sheet and rill erosion were quantitatively assessed, in a tropical mountainous river basin of the southern Western Ghats (India), viz., the KRB, using the coupled RUSLE-SEDD model. The soil erosion rate of the basin ranges from 0 to 796 t h^{-1} y^{-1} with a mean gross soil erosion of 9.3 t h^{-1} y^{-1}. The study shows that about 51% of the total area experiences slight erosion ($\leq$5 t/h y^{-1}), whereas roughly 5% of the catchment areas experience significant challenges due to severe and very severe erosion (>40 t h^{-1} y^{-1}). The estimated sediment yield of the basin ranges from 0 to 691 with a mean sediment yield of 6.3 t h^{-1} y^{-1} implying that the basin has significantly higher (67%) sediment transport efficiency. About 74% of the KRB yields sediment at a rate less than 5 t h^{-1} y^{-1}, whereas about 2% of the area generates very severe SY ($\geq$40 t h^{-1} y^{-1}). The critical erosion areas of the basin were delineated based on the HRUs of the basin and 72% of the catchment area belongs to very low and low-priority zones. We identified 71 HRUs having higher vulnerability to soil erosion requiring urgent attention towards implementation of appropriate soil erosion management strategies. The HRU-based analysis emphasized the significance of topography and vegetation in the soil erosion of the basin. This study demonstrates the suitability of the HRU-based prioritization of the critical erosion areas, which helps the watershed managers interfere at the specific locations within the river basin and implement efficient soil erosion management strategies. Although the present study examines the soil erosion severity of the region, we call for further studies addressing the significance of changing land use/ land cover and climate (with special reference to extreme weather events, such as floods and droughts) on soil erosion in the wake of changes in regional climate and socio-economic characteristics.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

References

Aiello, A., Adamo, M., Canora, F., 2015. Remote sensing and GIS to assess soil erosion with RUSLE3D and USPED at river basin scale in southern Italy. Catena 131, 174–185. https://doi.org/10.1016/j.catena.2015.04.003.

Ambili, V., Narayana, A.C., 2014. Tectonic effects on the longitudinal profiles of the Chaliyar River and its tributaries, southwest India. Geomorphology 217, 37–47. https://doi.org/10.1016/j.geomorph.2014.04.013.

Arnoldus, H.M.J., 1977. Methodology used to determine the maximum potential average annual soil loss due to sheet and rill erosion in Morocco. FAO Soils Bulletins (FAO).

Arnoldus, H.M.J., 1980. An approximation of the rainfall factor in the Universal Soil Loss Equation. An approximation of the rainfall factor in the Universal Soil Loss Equation., pp.127-132.

Avand, M., Nasiri Khiavi, A., Mohammadi, M., Tiefenbacher, J.P., 2023. Prioritizing sub-watersheds based on soil-erosion potential by integrating RUSLE and game-theory algorithms. Adv. Space Res. 72 (2), 471–487.

Batista, P.V., Laceby, J.P., Davies, J., Carvalho, T.S., Tassinari, D., Silva, M.L., Curi, N., Quinton, J.N., 2021. A framework for testing large-scale distributed soil erosion and sediment delivery models: Dealing with uncertainty in models and the observational data. Environ. Model. Softw. 137, 104961 https://doi.org/10.1016/j.envsoft.2021.104961.

Beasley, D.B., Huggins, L.F., Monke, A., 1980. ANSWERS: A model for watershed planning. Trans. ASAE 23 (4), 938–0944.

Bewket, W., Teferi, E., 2009. Assessment of soil erosion hazard and prioritization for treatment at the watershed level: case study in the Chemoga watershed, Blue Nile basin, Ethiopia. Land Degrad. Develop. 20 (6), 609–622. https://doi.org/10.1002/ldr.944.

Borah, D.K., Krug, E.C., Yoder, D., 2008. Watershed sediment yield. In Sedimentation Watershed sediment yield. In: Garcia, M.H. (Ed.), Sedimentation Engineering: Processes, Measurements, Modeling, and Practice, ASCE Manuals and Reports on Engineering Practice No. 110. American Society of Civil Engineers, Reston, pp. 827–858.

Chatterjee, S., Krishna, A.P., Sharma, A.P., 2014. Geospatial assessment of soil erosion vulnerability at watershed level in some sections of the Upper Subarnarekha river basin, Jharkhand, India. Environ. Earth Sci. 71, 357–374. https://doi.org/10.1007/s12665-013-2439-3.

Chinnasamy, P., Honap, V.U., 2023. Spatiotemporal variations in soil loss across the biodiversity hotspots of Western Ghats Region, India. J. Earth Syst. Sci. 132 (2), 90. https://doi.org/10.1007/s12040-023-02098-x.

FAO, I., 2015. Status of the World's Soil Resources (SWSR)–Main Report. Food and Agriculture Organization of the United Nations and Intergovernmental Technical Panel on Soils, Rome, Italy, 608.

Fayas, C.M., Abeysingha, N.S., Nirmanee, K.G.S., Samaratunga, D., Mallawatantri, A., 2019. Soil loss estimation using rusle model to prioritize erosion control in KELANI river basin in Sri Lanka. Int. Soil Water Conserv. Res. 7 (2), 130–137. https://doi.org/10.1016/j.iswcr.2019.01.003.

Fernandez, C., Wu, J.Q., McCool, D.K., Stöckle, C.O., 2003. Estimating water erosion and sediment yield with GIS, RUSLE, and SEDD. J. Soil Water Conserv. 58 (3), 128–136.

Ferro, V., Minacapilli, M., 1995. Sediment delivery processes at basin scale. Hydrol. Sci. J. 40 (6), 703–717. https://doi.org/10.1080/02626669509491460.

Ferro, V., Porto, P., Tusa, G., 1998. Testing a distributed approach for modelling sediment delivery. Hydrol. Sci. J. 43 (3), 425–442.

Fu, G., Chen, S., McCool, D.K., 2006. Modeling the impacts of no-till practice on soil erosion and sediment yield with RUSLE, SEDD, and ArcView GIS. Soil Tillage Res. 85 (1–2), 38–49. https://doi.org/10.1016/j.still.2004.11.009.

Haan, C.T., Barfield, B.J., Hayes, J.C., 1994. Design Hydrology and Sedimentology for Small Catchments. Elsevier.

Haregeweyn, N., Tsunekawa, A., Poesen, J., Tsubo, M., Meshesha, D.T., Fenta, A.A., Nyssen, J., Adgo, E., 2017. Comprehensive assessment of soil erosion risk for better land use planning in river basins: Case study of the Upper Blue Nile River. Sci. Total Environ. 574, 95–108. https://doi.org/10.1016/j.scitotenv.2016.09.019.

Hlaing, K.T., Haruyama, S., Aye, M.M., 2008. Using GIS-based distributed soil loss modeling and morphometric analysis to prioritize watershed for soil conservation in Bago river basin of Lower Myanmar. Front Earth Sci. China 2 (4), 465–478. https://doi.org/10.1007/s11707-008-0048-3.

Jain, M.K., Kothyari, U.C., 2000. Estimation of soil erosion and sediment yield using GIS. Hydrol. Sci. J. 45 (5), 771–786. https://doi.org/10.1080/02626660009492376.

Kanito, D., Bedadi, B., Feyissa, S., 2023. Sediment yield estimation in GIS environment using RUSLE and SDR model in Southern Ethiopia. Geomat. Nat. Haz. Risk 14 (1), 2167614. https://doi.org/10.1080/19475705.2023.2167614.

Kumar, S., Mishra, A., 2015. Critical erosion area identification based on hydrological response unit level for effective sedimentation control in a river basin. Water Resour. Manag. 29, 1749–1765. https://doi.org/10.1007/s11269-014-0909-3.

Li, L., Du, S., Wu, L., Liu, G., 2009. An overview of soil loss tolerance. Catena 78 (2), 93–99. https://doi.org/10.1016/j.catena.2009.03.007.

Masroor, M., Sajjad, H., Rehman, S., Singh, R., Rahaman, M.H., Sahana, M., Ahmed, R., Avtar, R., 2022. Analysing the relationship between drought and soil erosion using vegetation health index and RUSLE models in Godavari middle sub-basin, India. Geosci. Front. 13 (2), 101312 https://doi.org/10.1016/j.gsf.2021.101312.

Milliman, J.D., Syvitski, J.P., 1992. Geomorphic/tectonic control of sediment discharge to the ocean: the importance of small mountainous rivers. J. Geol. 100 (5), 525–544. https://doi.org/10.1086/629606.

Millward, A.A., Mersey, J.E., 1999. Adapting the RUSLE to model soil erosion potential in a mountainous tropical watershed. Catena 38 (2), 109–129. https://doi.org/10.1016/S0341-8162(99)00067-3.

Mirakhorlo, M.S., Rahimzadegan, M., 2020. Evaluating estimated sediment delivery by Revised Universal Soil Loss Equation (RUSLE) and Sediment Delivery Distributed (SEDD) in the Talar Watershed, Iran. Front. Earth Sci. 14, 50–62. https://doi.org/10.1007/s11707-019-0774-8.

Mitas, L., Mitasova, H., 1998. Distributed soil erosion simulation for effective erosion prevention. Water Resour. Res. 34 (3), 505–516. https://doi.org/10.1029/97WR03347.

Moore, I.D., Burch, G.J., 1986a. Modelling erosion and deposition: topographic effects. Trans. ASAE 29 (6), 1624–1630. https://doi.org/10.13031/2013.30363.

Moore, I.D., Burch, G.J., 1986b. Sediment transport capacity of sheet and rill flow: application of unit stream power theory. Water Resour. Res. 22 (8), 1350–1360. https://doi.org/10.1029/WR022i008p01350.

Moore, I.D., Wilson, J.P., 1992. Length-slope factors for the revised universal soil loss equation: simplified method of estimation. J. Soil Water Conserv. 47 (5), 423–428.

Morgan, R.P.C., Quinton, J.N., Smith, R.E., Govers, G., Poesen, J.W.A., Auerswald, K., Chisci, G., Torri, D., Styczen, M.E., 1998. The European Soil Erosion Model (EUROSEM): a dynamic approach for predicting sediment transport from fields and small catchments. Earth Surf. Proc. Land. 23 (6), 527–544.

Nearing, M.A., Foster, G.R., Lane, L.J., Finkner, S.C., 1989. A process-based soil erosion model for USDA-Water Erosion Prediction Project technology. Trans. ASAE 32 (5), 1587–1593. https://doi.org/10.13031/2013.31195.

Othman, A.A., Ali, S.S., Salar, S.G., Obaid, A.K., Al-Kakey, O. and Liesenberg, V., 2023. Insights for Estimating and Predicting Reservoir Sedimentation Using the RUSLE-SDR Approach: A Case of Darbandikhan Lake Basin, Iraq–Iran. Remote Sensing, 15 (3), p.697.

Ouadja, A., Benfetta, H., Porto, P., Mihoubi, M.K., Flanagan, D.C., Dehni, A., Talchabhadel, R., 2022. GIS and remote sensing integration for sediment performance assessment based on a RUSLE and sediment delivery ratio model in northwest Algeria. Arab. J. Geosci. 15 (5), 409. https://doi.org/10.1007/s12517-022-09502-8.

Pandey, A., Chowdary, V.M., Mal, B.C., 2007. Identification of critical erosion prone areas in the small agricultural watershed using USLE, GIS and remote sensing. Water Resour. Manag. 21, 729–746. https://doi.org/10.1007/s11269-006-9061-z.

Porto, P., Bacchi, M., Preiti, G., Romeo, M., Monti, M., 2022. Combining plot measurements and a calibrated RUSLE model to investigate recent changes in soil erosion in upland areas in Southern Italy. J. Soil. Sediment. 22 (3), 1010–1022.

Pradeep, G.S., Krishnan, M.N., Vijith, H., 2015. Identification of critical soil erosion prone areas and annual average soil loss in an upland agricultural watershed of Western Ghats, using analytical hierarchy process (AHP) and RUSLE techniques. Arab. J. Geosci. 8 (6), 3697–3711. https://doi.org/10.1007/s12517-014-1460-5.

Prasannakumar, V., Shiny, R., Geetha, N., Vijith, H., 2011a. Spatial prediction of soil erosion risk by remote sensing, GIS and RUSLE approach: a case study of Siruvani river watershed in Attapady valley, Kerala, India. Environ. Earth Sci. 64 (4), 965–972. https://doi.org/10.1007/s12665-011-0913-3.

Prasannakumar, V., Vijith, H., Geetha, N., Shiny, R., 2011b. Regional scale erosion assessment of a sub-tropical highland segment in the Western Ghats of Kerala, South India. Water Resour. Manage. 25 (14), 3715. https://doi.org/10.1007/s11269-011-9878-y.

Prasannakumar, V., Vijith, H., Abinod, S., Geetha, N., 2012. Estimation of soil erosion risk within a small mountainous sub-watershed in Kerala, India, using Revised Universal Soil Loss Equation (RUSLE) and geo-information technology. Geosci. Front. 3 (2), 209–215. https://doi.org/10.1016/j.gsf.2011.11.003.

Renard, K.G., Foster, G.R., 1983. Soil conservation: principles of erosion by water. In: Dregne HE, Willis WO (eds) Dryland agriculture. Agronomy Monograph 23, American Society of Agronomy, Crop Science Society of America, and Soil Science Society of America, Madison, Wisconsin, pp 155-176.

Renard, K.G., Foster, G.R., Weesies, G.A., McCool, D.K. and Yoder, D.C., 1997. Predicting soil erosion by water: a guide to conservation planning with the Revised Universal Soil Loss Equation (RUSLE) (Vol. 703). Washington, DC: United States Department of Agriculture.

Renard, K.G., Foster, G.R., Weesies, G.A., Porter, J.P., 1991. RUSLE: Revised universal soil loss equation. J. Soil Water Conserv. 46 (1), 30–33.

Richard, K., 1993. Sediment delivery and the drainage network. In: Beven, K., Kirkby, M. J. (Eds.), Channel Network Hydrology. John Wiley & Sons, Chichester, West Sussex, UK, pp. 222–254.

Saha, M., Sauda, S.S., Real, H.R.K., Mahmud, M., 2022. Estimation of annual rate and spatial distribution of soil erosion in the Jamuna basin using RUSLE model: a geospatial approach. Environ. Challen. 8, 100524 https://doi.org/10.1016/j.envc.2022.100524.

SCS (Soil Conservation Service) (1975) Urban hydrology for small watersheds. Technical release no. 55, Soil Conservation Service, United States Dept of Agric, Washington DC, USA.

Serbaji, M.M., Bouaziz, M., Weslati, O., 2023. Soil Water Erosion Modeling in Tunisia Using RUSLE and GIS Integrated Approaches and Geospatial Data. Land, 12(3), p.548.

Singh, S. and Kansal, M.L., 2023. Sub-basin prioritisation using RUSLE in a Mountainous River Basin of Uttarakhand (India). Environ., Develop. Sustain., pp.1-27.

Singh, G., Babu, R., Narain, P., Bhushan, L.S., Abrol, I.P., 1992. Soil erosion rates in India. J. Soil Water Conserv. 47 (1), 97–99.

Soman, K., 2002. Geology of Kerala. Geological Society of India, Bangalore.

Sourn, T., Pok, S., Chou, P., Nut, N., Theng, D., Prasad, P.V., 2022. Assessment of land use and land cover changes on soil erosion using remote sensing, GIS and RUSLE model: a case study of Battambang Province, Cambodia. Sustainability 14 (7), 4066.

SSO, 2007. Benchmark soils of Kerala. Soil Survey Organization, Government of Kerala, Thiruvananthapuram.

Sujatha, E.R., Sridhar, V., 2019. Mapping of erosion susceptibility using a weighted linear combination model: A case study of a hill sub-watershed in Kodaikkanal, Western Ghats, South India. Remote Sens. Appl.: Soc. Environ. 14, 34–45.

Tanyaş, H., Kolat, Ç., Süzen, M.L., 2015. A new approach to estimate cover-management factor of RUSLE and validation of RUSLE model in the watershed of Kartalkaya Dam. J. Hydrol. 528, 584–598. https://doi.org/10.1016/j.jhydrol.2015.06.048.

Thomas, J., Joseph, S., Thrivikramji, K.P., 2018a. Assessment of soil erosion in a monsoon-dominated mountain river basin in India using RUSLE-SDR and AHP. Hydrol. Sci. J. 63 (4), 542–560. https://doi.org/10.1080/02626667.2018.1429614.

Thomas, J., Joseph, S., Thrivikramji, K.P., 2018b. Assessment of soil erosion in a tropical mountain river basin of the southern Western Ghats. Geosci. Front. 9 (3), 893–906.

Thomas, J., Joseph, S., Thrivikramji, K.P., 2018c. Estimation of soil erosion in a rain shadow river basin in the southern Western Ghats. Int. Soil Water Conser. Res. 6 (2), 111–122.

Thomas, J., Prasannakumar, V., 2015. Comparison of basin morphometry derived from topographic maps, ASTER and SRTM DEMs: an example from Kerala, India. Geocarto Int. 30 (3), 346–364. https://doi.org/10.1080/10106049.2014.955063.

Tripathi, M.P., Panda, R.K., Raghuwanshi, N.S., 2003. Identification and prioritisation of critical sub-watersheds for soil conservation management using the SWAT model. Biosyst. Eng. 85 (3), 365–379. https://doi.org/10.1016/S1537-5110(03)00066-7.

USDA, 1978. Predicting Rainfall Erosion Losses. A Guide to Conservation Planning, Washington DC.

Van der Knijff, J.M., Jones, R.J.A. and Montanarella, L., 2000. Soil erosion risk assessment in Europe.

Wang, X., Yin, Z.Y., 1998. A comparison of drainage networks derived from digital elevation models at two scales. J. Hydrol. 210 (1–4), 221–241. https://doi.org/10.1016/S0022-1694(98)00189-9.

Williams, J.R., 1975. Sediment routing for agricultural watersheds. JAWRA J. Am. Water Resour. Assoc. 11 (5), 965–974.

Wischmeier, W.H. and Smith, D.D., 1978. Predicting rainfall erosion losses-a guide to conservation planning. Predicting rainfall erosion losses-a guide to conservation planning.

Wischmeier, W.H., Johnson, C.B., Cross, B.V., 1971. Soil erodibility nomograph for farmland and construction sites. J. Soil Water Conserv.